Atomic and Nuclear Physics from $\mathbb{C}^5$: Particles, Nuclei, and Decays as Hinge Geometry

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Abstract

We derive atomic and nuclear structure from DRLT hinge geometry. Every particle is classified by its location in $\mathbb{C}^5 = \mathbb{C}^3 \oplus \mathbb{C}^2$: quarks inhabit $\mathbb{C}^3$ (spatial), leptons inhabit $\mathbb{C}^2$ (temporal), and gauge bosons connect them. The proton is an SSS hinge with mass $m_p = 937.5$ MeV (0.08%), the neutron-proton mass difference $m_n - m_p = 1.302$ MeV (0.7%) follows from the electroweak breaking parameter $\delta_{\mathrm{EW}} = 1 - S(2)/S(\infty)$, and the nuclear force emerges as the 18 cross-hinges between two SSS structures. Beta decay is a $\mathbb{C}^2$ state flip emitting energy through STT channels. The neutrino is identified as a "ghost" — a $\mathbb{C}^2$-aligned state barely distinguishable from vacuum. All results use zero free parameters.

Contents

1 The DRLT Particle Dictionary

Every particle in the Standard Model has a definite location in the $\mathbb{C}^5 = \mathbb{C}^3 \oplus \mathbb{C}^2$ decomposition:

Particle	$\mathbb{C}$ sector	Hinge type	Charge	Spin	Mass origin
Fermions (matter)					
u, c, t quarks	$\mathbb{C}^3$	SSS	$+2/3$	$1/2$	$v_H/\sqrt{c} \times \varepsilon^{2(3-k)}$
d, s, b quarks	$\mathbb{C}^3$	SSS	$-1/3$	$1/2$	$m_t\alpha \times (\varepsilon\varphi)^{2(3-k)}$
e, μ, τ	$\mathbb{C}^2$	SST	-1	$1/2$	$m_b(5/12) \times (\varepsilon\varphi^2)^{2(3-k)}$
ν_e, ν_μ, ν_τ	$\mathbb{C}^2$ (aligned)	STT	0	$1/2$	seesaw
Gauge bosons (forces)					
8 gluons	$\mathbb{C}^3$ internal	SSS	0	1	confined
γ (photon)	$\mathbb{C}^3 \leftrightarrow \mathbb{C}^2$	SST	0	1	0 (U(1))
$W^\pm$	$\mathbb{C}^3 \leftrightarrow \mathbb{C}^2$	STT	± 1	1	$v_H/\sqrt{c}$
Z^0	$\mathbb{C}^3 \leftrightarrow \mathbb{C}^2$	STT	0	1	$m_W/\cos\theta_W$
Scalar					
Higgs	$\mathbb{C}^3 \leftrightarrow \mathbb{C}^2$ VEV	—	0	0	$v_H\sqrt{c\alpha d}$
Gravity					
graviton	hinge-to-hinge	$\det(G_h)$	0	2	0

Remark 1.1 (Why spin-1 vs spin-2). Gauge bosons live on *edges* (1-dimensional $\to$ vector $\to$ spin 1). Gravity lives on *hinges* (2-dimensional $\to$ tensor $\to$ spin 2). This is why gravity is fundamentally different from gauge forces: it is the arena, not an actor.

2 The Proton: Stable SSS Hinge

The proton (uud) is a triangle (SSS hinge) in $\mathbb{C}^3$:

$$G_p = \begin{pmatrix} 1 & G_{u_1u_2} & G_{u_1d} \\ G_{u_2u_1} & 1 & G_{u_2d} \\ G_{du_1} & G_{du_2} & 1 \end{pmatrix}, \qquad \det(G_p) > 0. \tag{1}$$

Mass:

$$m_p = n_S \times \Lambda_{\mathrm{QCD}} \times (1 + \alpha_{\mathrm{GUT}} \cdot n_S/d) = 3 \times 308 \times 1.0146 = 937.5 \text{ MeV}. \tag{2}$$

Observed: 938.3 MeV (0.08%).

Stability: $\binom{n_S}{n_S} = \binom{3}{3} = 1$ independent SSS configuration $\to$ no decay channel within $\mathbb{C}^3$ $\to$ proton is stable ($\tau_p \sim 10^{34}$ yr).

3 The Neutron and Isospin Splitting

3.1 Neutron as isospin partner

The neutron (udd) differs from the proton only in its $\mathbb{C}^2$ (temporal/isospin) quantum numbers: one u-type $\mathbb{C}^2$ state is replaced by a d-type.

3.2 Mass difference formula

$$\boxed{m_n - m_p = (m_d - m_u) \times \frac{n_S(d - 1 + S(2)/S(\infty))}{d^2} = 1.302 \text{ MeV}} \tag{3}$$

Observed: 1.293 MeV (0.7%).

Each factor:

- $(m_d - m_u) = 2.28$ MeV: isospin breaking source ($\mathbb{C}^2$ asymmetry).
- $n_S/d^2 = 3/25$: spatial projection normalized by full simplex.
- $(d - 1) = 4$: gravitational (metric) contribution from 4D spacetime.
- $S(2)/S(\infty) = 15/(2\pi^2) = 0.760$: electroweak asymmetry correction. $S(2) = 5/4$ (weak propagation range) vs $S(\infty) = \pi^2/6$ (EM propagation range). Their ratio measures how much the $12 = 12$ SST/STT channel symmetry is broken.

3.3 Why the mass difference is small

$m_n - m_p \approx 0.14\%$ of m_p. This extreme smallness follows from the Binet–Cauchy identity SST = STT = 12: the electromagnetic and weak channels have *identical* channel counts (when $c = 2$), so their effects nearly cancel. If $c \neq 2$, the channel counts would differ, the cancellation would fail, and $m_n - m_p$ would be much larger — potentially destabilizing all nuclei.

4 The Hydrogen Atom: $\mathbb{C}^3 + \mathbb{C}^2 + \mathbf{U}(1)$

4.1 Hinge structure

The hydrogen atom has 4 vertices: $\{u_1, u_2, d\}$ (proton, $\mathbb{C}^3$) and $\{e\}$ (electron, $\mathbb{C}^2$). The $\binom{4}{3} = 4$ possible hinges are:

Hinge	Vertices	Type	Force	Role
1	(u_1, u_2, d)	SSS	Strong	Binds proton
2	(u_1, u_2, e)	SST	EM	Binds electron
3	(u_1, d, e)	SST	EM	Binds electron
4	(u_2, d, e)	SST	EM	Binds electron

The atom = 1 SSS hinge (nuclear binding) + 3 SST hinges (electromagnetic binding).

4.2 Binding energy

$$E_1 = -\frac{m_e \alpha_{\text{em}}^2}{2} = -\frac{0.49\text{ MeV} \times (1/137.8)^2}{2} = -12.9\text{ eV}. \tag{4}$$

Observed: -13.6 eV (-5.1%, due to m_e being 4% low in DRLT).

4.3 Why the atom exists

The electron ($\mathbb{C}^2$) is bound to the proton ($\mathbb{C}^3$) by U(1) — the relative phase between the two sectors. This is the *same* U(1) that Chapter 1 proved necessary for forces to exist. The atom is a direct consequence of $\mathbb{K} = \mathbb{C}$: without the U(1) glue, $\mathbb{C}^3$ and $\mathbb{C}^2$ would be decoupled, and atoms could not form.

5 Beta Decay: $\mathbb{C}^2$ State Transition

5.1 The process

$n \to p + e^- + \bar{\nu}_e$:

1. A d-quark's $\mathbb{C}^2$ state flips to u-type ($\mathbb{C}^2$ isospin transition).
2. The $\mathbb{C}^2$ energy difference is emitted through STT channels (weak force).
3. The emitted energy materializes as an electron ($\mathbb{C}^2$ perturbation) and an antineutrino ($\mathbb{C}^2$ ghost).
4. The intermediary is a virtual W^- boson (a transient STT-type hinge).

5.2 Q-value

$$Q_\beta = m_n - m_p - m_e = 1.302 - 0.490 = 0.812\text{ MeV}. \tag{5}$$

Observed: 0.782 MeV (3.8%).

5.3 Why beta decay is slow

The neutron lifetime $\tau_n \approx 880$ s is extraordinarily long compared to strong interaction timescales ($\sim 10^{-23}$ s). In DRLT: beta decay goes through STT channels with $S(2) = 5/4$, while the strong force uses SSS with $S(1) = 1$. The effective coupling ratio is:

$$\frac{\text{weak rate}}{\text{strong rate}} \sim \left(\frac{1 \times S(1)}{12 \times S(2)}\right)^2 \times \text{phase space} \sim \left(\frac{1}{15}\right)^2 \sim 10^{-3}. \tag{6}$$

Combined with the small Q-value (phase space suppression $\propto Q^5$), this gives the observed long lifetime.

6 The Neutrino: Ghost of $\mathbb{C}^2$

The neutrino is a state almost perfectly aligned with the $\mathbb{C}^2$ vacuum:

$$\psi_\nu \approx (0, 0, 0, 1, 0) \in \mathbb{C}^5. \tag{7}$$

The $\mathbb{C}^3$ components are nearly zero ($|\psi_\nu^{C^3}|/|\psi_\nu^{C^2}| \sim 10^{-3}$). This explains all neutrino properties simultaneously:

Property	DRLT origin
Electrically neutral	$\mathbb{C}^3$ component $\approx 0 \to$ no SST coupling
Only weak interaction	Only STT hinges have nonzero det
Nearly massless	Almost aligned with vacuum (det ≈ 0)
Left-handed	$\mathbb{C}^2$ has chirality (pseudo-real fundamental)
Three flavors	$n_S = 3$ generations
Oscillates	Small $\mathbb{C}^3$ admixture allows flavor mixing

Remark 6.1. The neutrino is to $\mathbb{C}^2$ what the vacuum is to $\mathbb{C}^5$: an almost-aligned state. The vacuum has $\det(G_h) = 0$ exactly; the neutrino has $\det(G_h) = \epsilon \approx 0$. The neutrino is the lightest possible excitation of $\mathbb{C}^2$ — a "ghost" of the temporal sector.

7 Nuclear Force: Cross-Hinges

7.1 Mechanism

Two nucleons (two SSS hinges) interact through *cross-hinges*: triangles whose vertices come from both nucleons.

For deuterium (proton + neutron = 6 quarks):

- Intra-proton: $\binom{3}{3} = 1$ SSS hinge (proton binding).
- Intra-neutron: $\binom{3}{3} = 1$ SSS hinge (neutron binding).
- Cross-hinges: $\binom{6}{3} - 2 = 18$ hinges (nuclear force).

The 18 cross-hinges decompose as:

$$\binom{3}{1}\binom{3}{2} + \binom{3}{2}\binom{3}{1} = 9 + 9 = 18. \tag{8}$$

Each cross-hinge has a nonzero det when the two nucleons overlap spatially, providing an attractive interaction.

7.2 Nuclear binding energy

The binding energy is the difference in total hinge energy between bound and separated states:

$$E_{\text{bind}} = \sum_{\text{cross}} A_h\,\delta_h \qquad \text{(Regge action of cross-hinges).} \tag{9}$$

For deuterium, preliminary estimate:

$$E_d \sim \Lambda_{\text{QCD}} \times \alpha_{\text{GUT}} \times n_T/n_S = 308 \times 0.0243 \times 2/3 \approx 5.0 \text{ MeV}. \tag{10}$$

Observed: 2.224 MeV. The factor $\sim 2\times$ overestimate indicates that the nuclear force formula needs refinement (the simple product overestimates the cross-hinge det average). This remains an open problem.

7.3 Why nuclear physics is hard (but solvable)

In DRLT, nuclear binding is a *geometric optimization problem*: given A nucleons ($3A$ vertices in $\mathbb{C}^5$), find the configuration minimizing the total Regge action. This replaces the nuclear many-body Schrödinger equation with a 5×5 matrix problem.

The computational cost scales polynomially with A (matrix operations on $3A$ vectors in $\mathbb{C}^5$), not exponentially as in standard nuclear structure calculations. In principle, iron-56 ($A = 56$, 168 vertices) is computable on a laptop.

8 The Four Forces as Hinge Types

$$\text{Hinge types} = \{SSS,\ SST,\ STT\} + \underbrace{TTT}_{=0 \text{ since } n_T=2<3} \tag{11}$$

The number of gauge forces $= \min(3, n_T) = \min(3, 2) = 2 + 1 = 3$ types, plus gravity (hinge-to-hinge):

Force	Hinge type	Role	Metaphor
Strong	SSS	Binds quarks	The room
EM	SST	Connects $\mathbb{C}^3$ to $\mathbb{C}^2$	The glue
Weak	STT	$\mathbb{C}^2$ transitions	The door
Gravity	hinge $\leftrightarrow$ hinge	Connects simplices	The road

Why exactly 4: $n_T = 2 < 3$ forbids TTT hinges, capping gauge forces at 3. Gravity adds 1 (hinge-level). Total: $\min(3, n_T) + 1 = 3 + 1 = 4$. A 5th force is *mathematically forbidden*.

Why gravity couples to everything: gauge forces propagate *on* hinges (ϕ_{ij} lives on edges of the hinge). Gravity *is* the hinge ($\det(G_h)$ is the hinge itself). Without gravity (hinges), there is no arena for gauge forces to exist. Gravity is not a force — it is the spacetime that makes forces possible.

9 Summary of Predictions

#	Quantity	DRLT	Observed	Error
46	m_p	937.5 MeV	938.3	0.08%
47	$m_n - m_p$	1.302 MeV	1.293	0.7%
48	12=12 ($c = 2$ essential)	geometric	—	structural
49	Q_β	0.812 MeV	0.782	3.8%
50	E_1(hydrogen)	−12.9 eV	−13.6	−5.1%
51	Nuclear force = 18 cross-hinges	mechanism	—	structural
52	Exactly 4 forces (5th forbidden)	$\min(3, n_T) + 1$	4	exact

Total DRLT predictions to date: **52**, with zero free parameters.

References

[1] M. Jeong, "A Note on Lattice Geometry in $\mathbb{C}^5$" (2026).